



RunPod Architecture Plan for Monetchat Deployment

Overview

We are restructuring the Monetchat deployment to run fully on RunPod using a modular and scalable architecture. The goal is to separate concerns, improve performance, and make the system easier to manage and scale.

Current Situation

- Ollama (AI model server) is already deployed on a RunPod GPU pod
 - Monetchat application is currently running separately and needs proper integration
 - Some configurations still rely on local or mixed environments
-

Proposed Architecture

We will use a **multi-pod architecture inside RunPod**:

1. Ollama Pod (Already Running)

- Hosts AI models (e.g., qwen2.5:14b-instruct-q8_0)
 - Exposes /api/chat and /api/generate
 - Runs on GPU
 - Will be accessed internally (not publicly exposed)
-

2. Application Pod (New)

- Hosts Monetchat backend and frontend
- Includes:
 - Next.js app
 - API routes
 - Optional OpenAI-compatible proxy layer
- Runs on CPU or small GPU
- This will be the **only public-facing service**



3. Internal Communication (Key Change)

- Use **RunPod Global Networking** to connect:
App Pod → Ollama Pod (private connection)
- This removes dependency on public proxy URLs
- Improves:
 - latency
 - security
 - reliability

4. Optional: Network Volume (Later Phase)

- Used only if needed for:
 - persistent uploads
 - cached data
 - logs
- Not required for initial deployment

Request Flow

User → App Pod → Ollama Pod → Response → User

1. User interacts with Monetchat UI
2. App backend processes request
3. App calls Ollama internally via private network
4. Ollama returns AI response
5. App sends final response to user

Key Changes from Current Setup

- Remove all references to:
 - 127.0.0.1:11434
 - local Ollama instances
- Replace with:
 - internal RunPod Ollama endpoint
- Standardize API communication through one controlled layer



Benefits

1. Separation of Concerns

- App and AI model run independently
- Easier debugging and scaling

2. Scalability

- Can upgrade GPU (Ollama) without touching app
- Can scale app independently

3. Stability

- No dependency on local environments
- Fully cloud-based

4. Security

- Ollama is not publicly exposed
- Only app pod is accessible

Implementation Plan

Phase 1 – App Pod Setup

- Create new RunPod pod for application
- Deploy Monetchat code
- Configure environment variables

Phase 2 – Networking



- Enable Global Networking
- Connect app pod to Ollama pod
- Verify internal API communication

Phase 3 – Integration

- Update app to use:

```
OLLAMA_BASE_URL=http://<internal-ollama-host>:11434
```

-
- Test chat functionality end-to-end

Phase 4 – Stabilization

- Fix any latency or API issues
- Optimize streaming and responses

Phase 5 – Optional Improvements

- Add OpenAI-compatible proxy layer
- Introduce network volume if needed
- Add domain and SSL

Final Outcome

- Fully cloud-hosted Monetchat system on RunPod
 - Clean architecture with:
 - App layer
 - AI layer
 - Optional storage layer
 - Ready for scaling, production use, and future integrations
-